

Advanced C Programming - Assessment Questions

Question 1

What is the output of the following code ?

```
int main()
{
    int num = 0;

    int *p = num;

    num++;

    printf("%d
", *p);

    return 0;
}
```

- a. 1
- b. Segmentation fault
- c. 0
- d. Compiler error

Question 2

What is the output of the following code ?

```
int main()
{
    int num = -1;

    int *p = #

    printf("%d
", *p >> 1);

    return 0;
}
```

Answer: _____

Question 3

What is the output of the following code ?

```
int main()
{
    int num = -1;

    u8 *p = #

    printf("%d
", *p);

    return 0;
}
```

- a. -1
- b. Prints garbage value
- c. Segmentation fault
- d. Compiler error

Question 4

What is the output of the following code?

```
int main()
{
    int a[3] = {48, 65, 97};

    void *p = &a;

    printf("%c
", *p);

    return 0;
}
```

- a. Compiler error
- b. Segmentation fault
- c. 48
- d. 0

Question 5

What is the output of the following code ?

```
int main()
{
    int a[3] = {48, 65, 97};

    void *p = &a;

    printf("%c
", *(char *) (
(int *)p + 1));

    return 0;
}
```

Answer: _____

Question 6

What is the output of the following code ?

```
int main()
{
    char a[] = {48, 65, 97, 0};

    void *p = &a;

    printf("%s
", (char *)p + 1);

    return 0;
}
```

Answer: _____

Question 7

What is the output of the following code ?

```
int main()
{
    char a[] = "Hello";

    printf("%s
", a + 1);

    return 0;
}
```

Answer: _____

Question 8

What is the output of the following code ?

```
int main()
{
    char a[] = (char []) {"Hello"};

    printf("%s
", 1[a]++);

    return 0;
}
```

- a. Compiler error
- b. Segmentation fault
- c. Hello
- d. Hfllo

Question 9

What is the output of the following code ?

```
int main()
{
    char *a = "Hello";

    printf("%s
", a[1]++);

    return 0;
}
```

- a. Segmentation fault
- b. Hfllo
- c. Compiler error
- d. Hello

Question 10

For the below code, which statement will copy Hello in p ?

```
int main()

{

    char *p = malloc(6);

    // Statement

    return 0;

}
```

a. strcpy(*p, Hello);

b. strcpy("Hello", p);

c. strcpy(p, "Hello");

d. strcpy(p, Hello);

Question 11

What is the output of the following code ?

```
int main()
{
    char *a = calloc(6, 1);

    for (int i = 6; i--; )

        memset(a + 5 - i, 70 - i, 1);

    printf("%s
", a);

    return 0;
}
```

Answer: _____

Question 12

What is the output of the following code ?

```
int main()
{
    char *a = calloc(6, 1);

    for (int i = 6; i--; )

        memset(a + 5 - i, 53 - i, 1);

    printf("%s", a);

    bzero(a + 3 , 3);

    printf("%s", a);

    return 0;
}
```

Answer: _____

Question 13

Choose the correct statement among the following :

```
int (*p(void(*f[2])(int)))(char);
```

- a. f is a pointer which will collect integer & p is a function pointer which will collect the character
- b. f is an array of function pointers which will collect the character & p is an function pointer which will correct the integer
- c. f is a function pointer which will collect integer & p is a pointer which will collect the character
- d. f is an array of function pointers which will collect the integer & p is an function pointer which will correct the character

Question 14

Among the below options, choose the correct declaration for the following :

p is pointer to an function accepting an array of char returning a pointer to float

- a. float p(char *str[])
- b. float (*p(char *str[]))
- c. float *p (char *str[])
- d. float (*p)(char *str[])

Question 15

What is the output of the following code ?

```
int main()
{
    char x = 10;

    printf("%zu ", sizeof(x = 20));

    printf("%d
", x++);

    return 0;
}
```

Answer: _____

Question 16

What is the output of the following code ?

```
int main()
{
    int i;

    int *p1 = &i;

    int **p2 = &p1;

    for (**p2 = 0; *p1, i < 4; )

    {
        printf("%d", i);

        p2[0][0]++;
    }

    return 0;
}
```

Answer: _____

Question 17

What is the output of the following code ?

```
int main()
{
    int *p1 = NULL;

    int *p2;

    int a[4] = {10, 20, 30, 40};

    p2 = malloc(4 * sizeof(char));

    *(p2 + 3) = 10 ? *p1++: return 1;

    free(p1);

    printf("%d", p[3]);

    for (int i = 4; i--; )

        printf("%d ", *a++);

    return 0;
}
```

- a. Nothing will be printed
- b. Compiler error
- c. 40 30 20 10
- d. Segmentation fault

Question 18

What is the output of the following code ?

```
int main()

{

    char a[2][6];

    sscanf("HelloWorld", "%s", a[0]);

    sscanf("Blah", "%s", a[1]);

    printf("%s ", a[0]);

    printf("%s", a[1]);

    return 0;

}
```

Answer: _____

Question 19

What is the output of the following code ? (Assume we have included the header files required)

```
int add(int count, ...)

{

    int sum = 0;

    va_list ap;

    va_start(ap, count);

    sum += va_arg(ap, int);

    return sum;

}

int main()

{

    int ret = add(3, 4, 4, 2.5);

    printf("Sum is %#o", ret);

    return 0;

}
```

Answer: _____

Question 20

What is the output of the following code ?

```
int main()

{

    char a[2][3];

    void *p = a;

    sprintf((char *) p, "012345");

    for (int i = 0; i < 2; i++)

        for (int j = 0; j < 3; j++)

            printf("%u ", a[i][j]);

    return 0;

}
```

Answer: _____

Question 21

What is the output of the following code ?

```
int main()

{
char **p = malloc(2 * sizeof(char *));

for (int i = 0; i < 2 ; i++)

    p[i] = malloc(6);

strcpy( p[0] , "HELLO");

strcpy( p[1], "WORLD");

for (int i = 0; i < 2 ; i++)

    printf("%s ", p[i]);

return 0;
}
```

Answer: _____

Question 22

What is the output of the following code ?

```
int main()

{

    char i = 0;

    char *s[] = {

        "Array",

        "of",

        "Strings"

    };

    printf("%s", i++[s]);

    return 0;

}
```

Answer: _____

Question 23

Choose the operators for following operations :

Set a bit

Answer 1 Question 23

Choose...

>>

^

~

|

&

Clear and Get a bit

Answer 2 Question 23

Choose...

>>

^

~

|

&

Move a bit right

Answer 3 Question 23

Choose...

>>

^

~

|

&

Invert bits

Answer 4 Question 23

Choose...

>>

^

~

|

&

Toggle a bit

Answer 5 Question 23

Choose...

>>

^

~

|

&

Answer: _____

Question 24

What is the output of the following code ?

```
int main()

{

    int *p1 = (int *) 1000;

    static int *p2;

    printf("%d ", !(p1 && p2));

    printf("%d
", !(p2 || p1));

    return 0;

}
```

Answer: _____

Question 25

What is the output of the following code ?

```
int num;

int main()
{
    while (1)
    {
        func_1();

        num++;

        func_2();
    }

    printf("%d", num);

    return 0;
}
```

a. Compiler error

b. 1

c. 0

d. Infinite loop

Question 26

What is the output of the following code ?

```
int main()
{
    char i = 0;

    char *s[] = {
        "Array",
        "of",
        "Strings"
    };

    printf("%c
", s[0][i]++);

    return 0;
}
```

a. r

b. A

c. Compiler error

d. Segmentation fault

Question 27

What is the output of the following code ?

```
int main()

{

    int a[3][2] = {0, 1, 3, 4, 5, 6};

    int **p = a;

    for (int i = 0; i < 3; i++)

        for (int j = 0; j < 2; j++)

            printf("%d", (*(p + i))[j]);

    return 0;

}
```

- a. Compiler error
- b. Segmentation fault
- c. 013456 (compiles without warning)
- d. 013456 (compiles with warning)

Question 28

What is the output of the following code ?

```
int main()

{

    char *p1;

    p1 = 3 + "Hello World
", "Hell";

    printf("%s", p1);

    return 0;

}
```

Answer: _____

Question 29

What is the output of the following code ?

```
int num;

int main()

{

    num = 4;

    while (num--)

    {

        func_1();

        func_2();

    }

    printf("%d", num);

    return 0;

}
```

- a. 0
- b. 4
- c. Compiler error
- d. Infinite loop

Question 30

Replace the ? with correct options given below, so that we will get the output as 0x1881 :

```
int main()

{

    unsigned short x;

    unsigned char y = 0x81, z = 0x18;

    x = ? ;

    printf("%#x", x);

}
```

a. $(z \ll 8) \mid y$

b. $(z \gg 8) \& y$

c. $(y \ll 8) \& z$

d. $(y \gg 8) \mid z$

Question 31

What is the output of the following code ?

```
int main()

{
    unsigned char i, CP, DP, data[] = {0, 1, 2, 3};

    for (i = 0; i < 4; i++)

    {

        DP = data[i];

        CP = (DP & 0xF0) | (1 << i);

        printf("%x, %x
", DP, CP );

    }

}
```

Answer: _____

Question 32

What is the output of the following code ?

```
#define PORT 0xFF

#define LEDS PORT

int main()

{

    unsigned char p = LEDS;

    for (int i = 0; i < 7; i++)

    {

        p = p >> 1;

    }

    printf("%d", p);

}
```

Answer: _____

Question 33

What is the output of the following code ?

```
void foo(short num, char *buffer)

{
    for ( int i = 0; i < 4; )

    {

        *(buffer + i) = num % 10;

        buffer[i] += buffer[i] ;

        num /= 10;

        printf("%d ",*(buffer + i));

        i++;

    }

}

int main()
{
    char buffer[] = {1, 2, 3, 4} ;

    foo(2, buffer);

}
```

Answer: _____

Question 34

What is the output of the following code ?

```
int main()
{

    int *p1 = (int *) 1000;

    int *p2 = (int *) 1020;

    printf("%ld ", p1 - p2);

    printf("%ld ", p2 - p1);

    puts("
");

    return 0;

}
```

Answer: _____

Question 35

Which operator would you use to test a byte for its truthness?

a. |

b. ^

c. &

d. &&